What I Learned from Assignment 2

Working through this lab helped me become more familiar with Python and some of the tools commonly used in machine learning. Since I am still learning Python, GitHub, Google Colab, and machine learning concepts at the same time, there were several things that were new to me. Even some of the smaller parts of the code helped me understand how Python programs are structured.

One of the first things I learned was how to comment out text in Python so that it is ignored when the code is being run. I also learned how to import Python libraries and give them nicknames, or aliases, so they can be used more conveniently later. For example, pandas can be imported as pd and NumPy can be imported as np. I learned how to print text to the console using Python and how to print information such as the current version of a library. These were relatively simple concepts, but they helped me become more comfortable reading and understanding Python code.

I also learned about the Iris dataset, which is a commonly used beginner dataset in machine learning. I learned how to import a dataset and store it in a variable. This helped me understand that a variable does not necessarily have to contain a single number or word. It can also contain a much larger collection of data that can then be accessed and manipulated using Python.

Another important thing I learned was how pandas can be used to display and work with data in the form of rows and columns. I learned how to create a DataFrame and how to see information about its size and contents. Being able to view the data in a table made it much easier to understand what the dataset actually contained. I also learned how to access individual columns and perform basic calculations on them.

The lab also introduced me to the for loop in Python. I learned that a for loop can repeat a section of code for each item in a collection. This became especially useful when creating the scatter plot because the code could repeat the same process for each species without having to write the same commands three separate times.

Finally, I learned that Python can perform simple shortcuts that make code more efficient. For example, using "=" * 40 repeats a character without requiring me to type it forty times. While this is a small feature, it showed me that Python has many ways to make code shorter and easier to manage.

The main thing that I struggled with really, was reading everything since it was presented all at once. But this is only the first assignment of many. I'll be sure to teach myself even more so that I am better prepared for the next assignment.